Water Quality Prediction

Problem & Scope

Predicting the CCME Water Quality Index from physical and chemical measurements

Naive baseline, Random Forest, XGBoost | CSAI (AML / DL) Project 18

- Goal: predict the continuous water quality index (**CCME_Values**) from routine physical and chemical measurements
- Task type: **regression** (observed range 31.304 to 100.0)
- Compares a Naive baseline, Random Forest and XGBoost on one shared, leakage-safe split
- Every cleaning, feature and tuning decision is logged with its reason

Scope. All three models are scored on the **same** test set via the same **evaluate()** call. Both tree models expose the same training-function signature.

2,827,977
rows in the raw dataset (14 columns)

5
countries: England, USA, Ireland, China, Canada

61,689
unique monitoring sites (Area) in the raw data

Dataset

Data Analysis

Harmonized global surface-water monitoring dataset, 5 countries

Rows by country (raw data)

Country	Rows
England	2,129,198
USA	413,814
Ireland	235,019
China	45,997
Canada	3,949

What is in the data

- 14 columns: **Country**, **Area**, **Waterbody Type**, **Date**; 8 measurement columns (Ammonia, BOD, Dissolved Oxygen, Orthophosphate, pH, Temperature, Nitrogen, Nitrate); **CCME_Values** and **CCME_WQI**
- Date coverage: 2 January 1940 to 17 December 2023
- Waterbody Type: 12 classes; River 65.51%, Effluent 21.27%, Lake 5.43%
- England is 75% of rows, so country shift matters for any split
- Raw data is never edited in place

2,818,006
rows x 30 columns after cleaning

85.05
mean CCME (median 90.60)

17.65
standard deviation of the target

45 MB
clean.parquet, committed (plain git, no LFS)

Data Cleaning in Detail

Cleaning

What was checked, what was flagged, what was removed

Missing values	Zero missing values reported. No NaN removal and no imputation during cleaning: imputation lives in the preprocessing pipeline and is fitted on **train only**.
Upstream median fill	Modal value check: 5 of 8 columns carry their exact median as the most frequent value. Dissolved oxygen 10.2 (31.60% of rows), nitrate 4.5 (25.33%), BOD 2.7 (24.28%), pH 7.78 (12.98%), temperature 11.46 (11.02%). Flagged with one boolean per column plus **n_imputed_upstream** (0 to 8).
Units and limits	Units verified per column (mg/l, pH units, °C); no mixed units. 5,508 rows contain a '<', all inside Area site names; all 8 measurement columns parse as pure floats, so no detection-limit values.
Impossible values	Flagged, not deleted: 505 pH readings outside 0 to 14, 15 negative concentrations, 7,284 temperatures above 40 °C (**flag_any_impossible**: 7,676 rows).
Duplicates	9,971 exact duplicate rows (0.35%) removed: 2,827,977 to 2,818,006 rows.
Output	2,818,006 rows x 30 columns written to **clean.parquet** (45 MB). Column names kept verbatim.

Exploratory Data Analysis

EDA

Nine findings that change modelling decisions

35.92%

of rows have CCME_Values exactly 100.0: the target is bounded at its ceiling, with a large point mass on the bound

20.9% to 84.5%

share of rows at the ceiling, England to USA: target shape differs by country

11.98

skewness (ammonia), kurtosis 222; log1p brings skewness to 3.35

-0.917

Spearman correlation of orthophosphate with the target

about 0

dissolved oxygen correlation with the target (Pearson -0.031)

2.04 vs 0.04

suspect measurements per row: England vs every other country (at most 0.04)

22%

of rows are Effluent (21.3%) or Sewage (0.8%): mean target 66.6 vs 90.0 for rivers

82%

of rows fall in 2000 to 2019; England's records begin in 2000

2.0 to 16.6%

of rows per column marked by the IQR rule: outliers not removed, cleaning flags kept

Target and Leakage

Target Definition

CCME Water Quality Index (CCME_Values)

$CCME_Values = 100 - sqrt(F1^2 + F2^2 + F3^2) / 1.732$

- **F1 (Scope):** percentage of variables that do not meet their guideline
- **F2 (Frequency):** percentage of tests that do not meet guidelines (set equal to F1 here)
- **F3 (Amplitude):** amount by which failed values exceed their guidelines
- Dividing by 1.732 (square root of 3) normalizes the maximum distance, 173.2, to 100

Index bands

95 - 100	Excellent
80 - 94	Good
65 - 79	Fair
45 - 64	Marginal
0 - 44	Poor

The 8 physical/chemical measurements reconstruct CCME_Values with 0.000000 error.

Ammonia, BOD, Dissolved Oxygen, Orthophosphate, pH, Temperature, Nitrogen, Nitrate. Therefore, the 8/8 model is a **surrogate/emulator** of the CCME formula rather than independent discovery of water-quality physics (labelled **surrogate_emulator_benchmark**).

Leakage controls

- **CCME_WQI** is excluded: it is the target cut into 5 classes, so it is direct label leakage
- **Area** and raw **Date** are excluded from the features; only sin_day, cos_day and month are kept
- Partial-measurement scenarios (7/8, 6/8, 4/8) test machine learning usefulness when measurements are unavailable

Feature Engineering

Features

Derived features from the 8 measurements and the Date

Ratios (5)	**N_P_ratio** = Nitrogen / Orthophosphate | **Inorganic_N_ratio** = Ammonia / Nitrate | **Organic_N_estimate** = Nitrogen - (Nitrate + Ammonia), clipped at 0 | **Organic_N_fraction** = Organic_N_estimate / Nitrogen (0 to 1) | **BOD_DO_ratio** = BOD / Dissolved Oxygen
Log transforms (5)	**log_Ammonia, log_BOD, log_Orthophosphate, log_Nitrate, log_Nitrogen**: monotonic transforms for positively skewed, heavy-tailed measurements
Temporal (3)	**sin_day, cos_day, month**: annual seasonality from Date (raw Date and Area are not used)
Safeguards	Negative concentrations clipped at 0 before ratios and logs; denominators protected by EPSILON (0.001) or a Dissolved Oxygen floor (0.01); NaN propagates, no imputation here (it lives in preprocessing); sub-zero temperature kept as a valid reading; BOD:COD excluded (COD is not in the dataset)
Ablation masking	Derived features are dropped automatically when their parent measurements are masked in the 7/8, 6/8 and 4/8 scenarios

10 to 23
features, raw vs engineered (XGBoost)

0.1517 to 0.0999
MAE, raw vs engineered (XGBoost, test)

0.4005 to 0.3282
RMSE, raw vs engineered (XGBoost, test)

Method and Fairness Rules

Pipeline & Models

One shared split, train-only preprocessing, test set touched once

1 Raw data | 2 Clean | 3 Shared split | 4 Preprocess | 5 Train 3 models | 6 Evaluate once

Fairness rules (every model)

- One split, **SEED = 42**, assignments saved to disk
- Split by **Area** (monitoring station), stratified by Country: no station appears in two partitions
- Preprocessing and aggregates fitted on train only
- Tuning on validation only
- Test set used exactly once
- All experiments use the same fixed Area-grouped test set

Models and training sizes

- **Naive baseline:** median predictor, the floor to beat
- **Random Forest:** search over trees, depth, min leaf size, features per split; **100,000** training samples
- **XGBoost:** Optuna search with early stopping; **1,986,397** training samples

Shared split (70 / 15 / 15)

Partition	Rows	Stations
Train	1,986,397	43,382
Validation	415,805	9,153
Test	415,804	9,154

China: all 45,977 rows come from one station (Hou Bay), allocated to train, so China has 0 test samples.

MAE — mean absolute error

RMSE — penalizes large errors more

R^2 — variance explained

Training sizes. Random Forest uses a fixed 100k computational budget; XGBoost uses the full training partition.

Results and Team

Evaluation & Docs

Test set, same split, same evaluate() call

Test results

Model	MAE	RMSE	R^2
Naive baseline (median)	13.6047	19.4916	-0.0923
Random Forest (100k rows)	0.4078	0.9197	0.9976
XGBoost (full, engineered)	0.1000	0.3282	0.9997

Ablation (XGBoost, test)

Configuration	MAE	RMSE
Raw features (10)	0.1517	0.4005
Raw + engineered (23)	0.0999	0.3282

SHAP. Orthophosphate, log_Orthophosphate and Ammonia carry the largest mean absolute SHAP values.

Partial measurements (XGBoost, test)

Available	Removed	RMSE	R^2
8/8	none	0.3282	0.9997
7/8	Orthophosphate	7.5966	0.8341
6/8	Orthophosphate + BOD	8.0451	0.8139
4/8	Orthophosphate + BOD + Ammonia + Nitrogen	12.1898	0.5728

Performance degrades sharply as measurements become unavailable, demonstrating that the strong 8/8 performance depends on access to the measurements used by the CCME calculation.

Team Tajuddin: data and EDA | Parthraj: target, split, evaluation | Himanshu: features, Random Forest | Jivit: XGBoost | Satya: metrics, baseline, report and deck